Lab Exercise 11- Deployments with Rolling Update and Recreate Strategies

Understand how to use the rolling update and recreate strategies for deploying applications using Kubernetes Deployments.

Step 1: Create a Deployment with Rolling Update Strategy

Create a YAML file for the deployment:

Create a file named **nginx-deployment-rolling.yaml** with the following content:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment-rolling
spec:
  replicas: 10
  minReadySeconds: 10
  selector:
    matchLabels:
      app: web
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 5
  template:
    metadata:
      labels:
```

```
      app: web
    spec:
      containers:
      - name: nginx
        image: hkshitesh/kubedemo:1.0
        ports:
        - containerPort: 80
```

Apply the deployment:

```
kubectl apply -f nginx-deployment-rolling.yaml ; watch "kubectl get rs -o wide"
```

Verify the deployment:

```
kubectl get deployments
kubectl get pods -l app=web
```

Update the deployment to a new image:

```
kubectl set image deployment/nginx-deployment-rolling nginx=  hkshitesh/kubedemo:2.0
```

Monitor the rolling update:

```
kubectl rollout status deployment nginx-deployment-rolling
```

Verify the updated pods:

```
kubectl get pods -l app=web -o wide
```

Step 2: Create a Deployment with Recreate Strategy

Create a YAML file for the deployment:

Create a file named **nginx-deployment-recreate.yaml** with the following content:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment-recreate
spec:
  replicas: 10
  selector:
    matchLabels:
      app: web
  strategy:
    type: Recreate
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: nginx
        image: nginx:  hkshitesh/kubedemo:1.0
        ports:
        - containerPort: 80
```

Apply the deployment:

```
kubectl apply -f nginx-deployment-recreate.yaml ; watch "kubectl get rs -o wide"
```

Verify the deployment:

```
kubectl get deployments
kubectl get pods -l app=nginx-recreate
```

Update the deployment to a new image:

```
kubectl set image deployment/nginx-deployment-recreate nginx=nginx:1.21.1
```

Monitor the update:

```
kubectl rollout status deployment nginx-deployment-recreate
```

Verify the updated pods:

```
kubectl get pods -l app=nginx-recreate -o wide
```

Step 3: Clean Up

Delete the deployments:

```
kubectl delete deployment nginx-deployment-rolling
kubectl delete deployment nginx-deployment-recreate
```

Verify that all resources are cleaned up:

```
kubectl get deployments
```

kubectl get pods -l app=nginx

kubectl get pods -l app=nginx-recreate

```
C:\Users\NK>(echo apiVersion: apps/v1 & echo kind: Deployment & echo metadata: & echo   name: nginx-deployment-rolling & echo spec: & echo   replicas: 10 &
echo   minReadySeconds: 10 & echo   selector: & echo     matchLabels: & echo       app: web & echo   strategy: & echo     type: RollingUpdate & echo     rol
lingUpdate: & echo       maxUnavailable: 1 & echo       maxSurge: 5 & echo   template: & echo     metadata: & echo       labels: & echo         app: web & e
cho     spec: & echo       containers: & echo       - name: nginx & echo         image: hkshitesh/kubedemo:1.0 & echo         ports: & echo         - contai
nerPort: 80) > nginx-deployment-rolling.yaml

C:\Users\NK>type nginx-deployment-rolling.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment-rolling
spec:
  replicas: 10
  minReadySeconds: 10
  selector:
    matchLabels:
      app: web
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 5
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: nginx
        image: hkshitesh/kubedemo:1.0
        ports:
        - containerPort: 80

C:\Users\NK>kubectl apply -f nginx-deployment-rolling.yaml
deployment.apps/nginx-deployment-rolling created

C:\Users\NK>kubectl get deployments
NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment-rolling    0/10    10           0           3s

C:\Users\NK>kubectl get pods -l app=web
```

```
C:\Users\NK>kubectl get pods -l app=web
NAME                                      READY   STATUS              RESTARTS   AGE
my-pod                                    1/1     Running             0          9m16s
nginx-deployment-rolling-85cf8767cc-2ks2l   0/1     ContainerCreating   0          7s
nginx-deployment-rolling-85cf8767cc-8rcvd   0/1     ContainerCreating   0          7s
nginx-deployment-rolling-85cf8767cc-k4bgv   0/1     ContainerCreating   0          7s
nginx-deployment-rolling-85cf8767cc-mdzdm   0/1     ContainerCreating   0          7s
nginx-deployment-rolling-85cf8767cc-nkrt2   0/1     ContainerCreating   0          7s
nginx-deployment-rolling-85cf8767cc-p64hn   0/1     ContainerCreating   0          7s
nginx-deployment-rolling-85cf8767cc-rgj7l   0/1     ContainerCreating   0          7s
nginx-deployment-rolling-85cf8767cc-s64gd   0/1     ContainerCreating   0          7s
nginx-deployment-rolling-85cf8767cc-x9f2g   0/1     ContainerCreating   0          7s
nginx-deployment-rolling-85cf8767cc-zp8xc   0/1     ContainerCreating   0          7s

C:\Users\NK>kubectl set image deployment/nginx-deployment-rolling nginx=hkshitesh/kubedemo:2.0
deployment.apps/nginx-deployment-rolling image updated

C:\Users\NK>kubectl rollout status deployment nginx-deployment-rolling
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
Waiting for deployment "nginx-deployment-rolling" rollout to finish: 6 out of 10 new replicas have been updated...
```

```
C:\Users\NK>(echo apiVersion: apps/v1 & echo kind: Deployment & echo metadata: & echo   name: nginx-deployment-recreate & echo spec: & echo   replicas: 10 &
 echo   selector: & echo     matchLabels: & echo       app: web & echo   strategy: & echo     type: Recreate & echo   template: & echo     metadata: & echo
      labels: & echo         app: web & echo     spec: & echo       containers: & echo       - name: nginx & echo         image: hkshitesh/kubedemo:1.0 & ec
ho         ports: & echo         - containerPort: 80) > nginx-deployment-recreate.yaml

C:\Users\NK>type nginx-deployment-recreate.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment-recreate
spec:
  replicas: 10
  selector:
    matchLabels:
      app: web
  strategy:
    type: Recreate
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: nginx
        image: hkshitesh/kubedemo:1.0
        ports:
        - containerPort: 80

C:\Users\NK>kubectl apply -f nginx-deployment-recreate.yaml
deployment.apps/nginx-deployment-recreate created

C:\Users\NK>kubectl get deployments
NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment-recreate   8/10    10           8           4s
nginx-deployment-rolling    10/10   10           10          110s

C:\Users\NK>kubectl get pods -l app=web
NAME                                        READY   STATUS    RESTARTS   AGE
my-pod                                      1/1     Running   0          11m
nginx-deployment-recreate-85cf8767cc-25mjw  1/1     Running   0          10s
nginx-deployment-recreate-85cf8767cc-dtks2  1/1     Running   0          10s
```

```
C:\Users\NK>kubectl rollout status deployment nginx-deployment-recreate
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 7 of 10 updated replicas are available...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 8 of 10 updated replicas are available...
Waiting for deployment "nginx-deployment-recreate" rollout to finish: 9 of 10 updated replicas are available...
deployment "nginx-deployment-recreate" successfully rolled out

C:\Users\NK>kubectl get pods -l app=web -o wide
NAME                                        READY   STATUS    RESTARTS   AGE   IP            NODE       NOMINATED NODE   READINESS GATES
my-pod                                      1/1     Running   0          11m   10.244.0.2    minikube   <none>           <none>
nginx-deployment-recreate-8587c95656-7l57q  1/1     Running   0          8s    10.244.0.42   minikube   <none>           <none>
nginx-deployment-recreate-8587c95656-8pk46  1/1     Running   0          8s    10.244.0.41   minikube   <none>           <none>
nginx-deployment-recreate-8587c95656-cs8d7  1/1     Running   0          8s    10.244.0.40   minikube   <none>           <none>
nginx-deployment-recreate-8587c95656-dwpxc  1/1     Running   0          8s    10.244.0.47   minikube   <none>           <none>
nginx-deployment-recreate-8587c95656-n8kcp  1/1     Running   0          8s    10.244.0.44   minikube   <none>           <none>
nginx-deployment-recreate-8587c95656-psbpk  1/1     Running   0          8s    10.244.0.45   minikube   <none>           <none>
nginx-deployment-recreate-8587c95656-tmdqv  1/1     Running   0          8s    10.244.0.39   minikube   <none>           <none>
nginx-deployment-recreate-8587c95656-vgzpk  1/1     Running   0          8s    10.244.0.43   minikube   <none>           <none>
nginx-deployment-recreate-8587c95656-x9mzf  1/1     Running   0          8s    10.244.0.48   minikube   <none>           <none>
nginx-deployment-recreate-8587c95656-z9cwz  1/1     Running   0          8s    10.244.0.46   minikube   <none>           <none>
nginx-deployment-rolling-8587c95656-5sp2w   1/1     Running   0          66s   10.244.0.27   minikube   <none>           <none>
nginx-deployment-rolling-8587c95656-7d8rl   1/1     Running   0          2m    10.244.0.22   minikube   <none>           <none>
nginx-deployment-rolling-8587c95656-7sk5s   1/1     Running   0          69s   10.244.0.26   minikube   <none>           <none>
nginx-deployment-rolling-8587c95656-9kptw   1/1     Running   0          2m    10.244.0.19   minikube   <none>           <none>
nginx-deployment-rolling-8587c95656-b6skk   1/1     Running   0          2m    10.244.0.23   minikube   <none>           <none>
nginx-deployment-rolling-8587c95656-gzvcx   1/1     Running   0          2m    10.244.0.24   minikube   <none>           <none>
nginx-deployment-rolling-8587c95656-jlbl8   1/1     Running   0          2m    10.244.0.21   minikube   <none>           <none>
nginx-deployment-rolling-8587c95656-kht2x   1/1     Running   0          64s   10.244.0.28   minikube   <none>           <none>
nginx-deployment-rolling-8587c95656-r9ct8   1/1     Running   0          71s   10.244.0.25   minikube   <none>           <none>
nginx-deployment-rolling-8587c95656-zsq6m   1/1     Running   0          2m    10.244.0.20   minikube   <none>           <none>

C:\Users\NK>kubectl delete deployment nginx-deployment-rolling
deployment.apps "nginx-deployment-rolling" deleted from default namespace

C:\Users\NK>kubectl delete deployment nginx-deployment-recreate
deployment.apps "nginx-deployment-recreate" deleted from default namespace

C:\Users\NK>kubectl get deployments
No resources found in default namespace.

C:\Users\NK>kubectl get pods -l app=web
NAME     READY   STATUS    RESTARTS   AGE
```